Supplementary material

qPCRtools: a cross-platform C++ desktop application for qPCR data analysis with validated equivalence to its R implementation

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1 Analysis output on the bundled example data

Figure S1 shows the two analyses discussed in the main text, produced from the application's numerical output by the plotting script shipped in the repository (`code/validation/plot_examples.py`, matplotlib), which gives finer control over the layout than the in-interface figure export. The per-group means, standard deviations and p-values were independently reproduced without the application or the R package by `code/validation/check_example_stats.R`.

2 Numerical equivalence to the R package

Both implementations were run on identical input files for every shared method—$\Delta\Delta C_t$ (`CalExp2ddCt`), the standard curve (`CalCurve`) and the efficiency-based RqPCR method (`CalExpRqPCR`)—and every resulting quantity was compared pairwise (Figure S2, Table S1).

Table 1: **Table S1.** Equivalence of C++ and R results across all shared methods on identical inputs. Relative difference is $|R - \text{C++}|/|R|$.

Method	Compared quantities	Pairs	Max. rel. diff.
$\Delta\Delta C_t$	expression, SD	6	3.8×10^{-15}
Standard curve	slope, intercept, R^2, efficiency	16	2.5×10^{-3}
RqPCR	expression, SE	16	2.1×10^{-15}

The residual 2.5×10^{-3} for the standard curve is attributable to the R package rounding its tabulated output to two to four decimal places, since the underlying C_q inputs are identical.

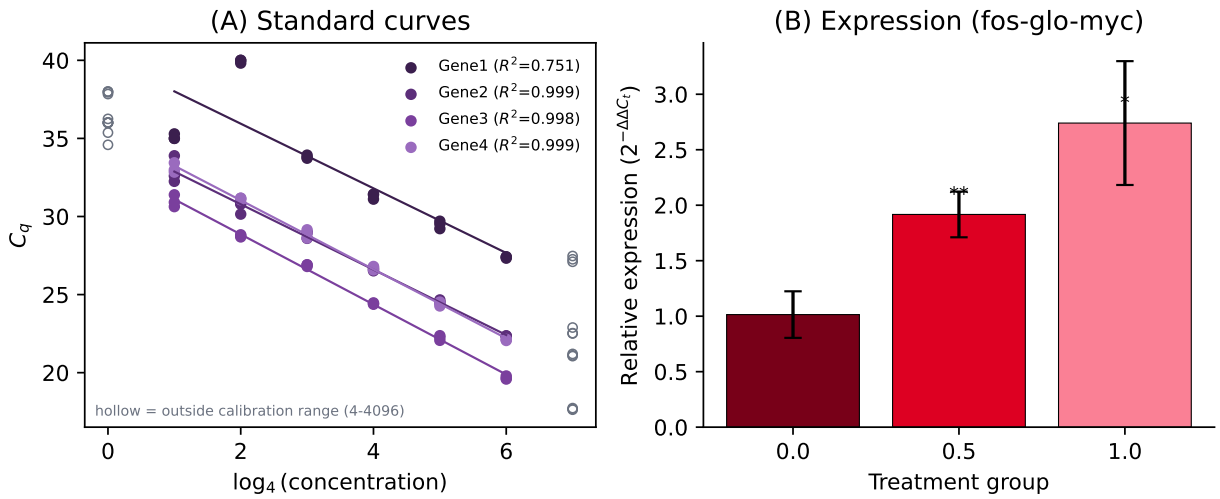


Figure 1: **Figure S1.** Analysis output on real data. (A) Standard curves for four genes over a four-fold dilution series, with fitted regressions and R^2. The abscissa is $\log_4$ concentration (dilution steps), matching the regression, and hollow symbols mark wells outside the calibrated range (4–4096) that are excluded from the fit. (B) Relative expression of *fos-glo-myc* across treatment groups ($\Delta\Delta C_t$ method, calibrated to the control group), with significance labels. Reproduced from `figures/fig2_results_v2.pdf`.

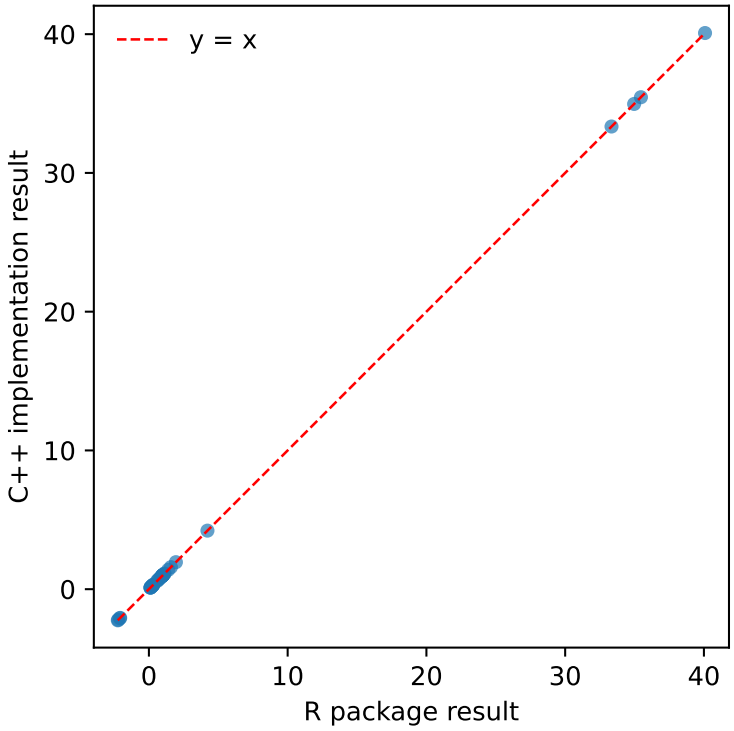


Figure 2: **Figure S2.** Numerical equivalence of the C++ implementation and the R package. Each C++ result is plotted against the corresponding R result across all shared methods; all points fall on the line $y = x$.